

Task 4 Report — Responsible AI & Model Interpretation

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1. Objective

Analyze the fairness, bias, and explainability of the Random Forest classifier built in Task 1 (Titanic survival prediction) using SHAP and LIME techniques, and propose practical mitigation steps.

2. Tools Used

- **SHAP** (SHapley Additive exPlanations) — for global and local model explainability
 - **LIME** (Local Interpretable Model-agnostic Explanations) — for individual prediction explanation
 - **Pandas** — for group-wise bias analysis across sensitive attributes
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3. SHAP Analysis — Global Feature Importance

The SHAP summary plot reveals which features most influence the model's predictions across all passengers:

Rank	Feature	Impact
1	Sex	Highest — being female strongly increases survival probability
2	Pclass	Higher class = higher survival
3	Fare	Higher fare = higher survival
4	Age	Moderate impact
5	Embarked	Small impact
6	SibSp	Small impact
7	Parch	Least impact

Interpretation: The model heavily relies on `sex` as the primary predictor — this reflects real historical data ("women and children first") but also means the model has learned a strong gender-based decision rule.

4. SHAP Waterfall — Local Explanation (Passenger 0)

For Passenger 0 (male, died — prediction: 0.22 survival probability):

Feature	SHAP Value	Effect
Sex (male)	-0.12	Strongly pushed prediction DOWN
Pclass (3rd)	-0.07	Pushed prediction DOWN
SibSp	+0.07	Slight positive push
Embarked	-0.04	Pushed DOWN
Age	-0.02	Slight negative
Parch	+0.02	Slight positive
Fare	-0.01	Minimal effect

Baseline ($E[f(X)]$): 0.385 → Final prediction: 0.22 (correctly predicted as "Died")

5. LIME — Local Explanation

LIME independently confirmed SHAP's finding: `sex_enc` (being male) contributed -0.35 to the prediction — by far the largest single factor. Being in 3rd class and a non-Southampton embarkation port further reduced survival probability.

6. Bias Analysis

6.1 Bias by Sex

Group	Actual Rate	Survival Predicted Rate	Survival Difference
Female	73.8%	75.4%	+1.6% (slight over-prediction)
Male	20.3%	13.6%	-6.7% (significant under-prediction)

Finding: The model under-predicts male survival by 6.7 percentage points. It is systematically more pessimistic about male passengers than reality warrants.

6.2 Bias by Passenger Class

Class	Actual Survival Rate	Predicted Survival Rate	Difference
1st	55.6%	55.6%	0% (perfect)
2nd	58.8%	52.9%	-5.9%
3rd	24.0%	19.0%	-5.0% (under-prediction)

Finding: The model is fair for 1st class passengers but consistently under-predicts survival for 2nd and 3rd class passengers. This reflects **socioeconomic bias** — the model penalises lower-class passengers more than historical data justifies.

7. Root Cause of Bias

The bias is not introduced by the model itself — it is **inherited from the training data**. The Titanic dataset reflects a real historical event where gender and social class dramatically influenced who survived. A model trained on this data will learn and replicate these historical inequalities.

8. Mitigation Recommendations

- 1. Re-weighting samples:** Assign higher training weights to under-represented groups (male, 3rd class) so the model pays more attention to their patterns.
 - 2. Fairness constraints:** Use fairness-aware algorithms (e.g., `fairlearn` library) that add constraints to equalise false negative rates across groups during training.
 - 3. Separate threshold tuning:** Instead of a single 0.5 decision threshold for all passengers, use group-specific thresholds calibrated to equalise prediction rates across sex and class groups.
 - 4. Remove or de-weight sensitive attributes:** Experiment with removing `sex` as a direct feature and using only proxy-free features, though this may reduce accuracy.
 - 5. Regular bias auditing:** In any production ML system, regularly monitor model predictions across sensitive groups after deployment — bias can worsen as data distributions shift over time.
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9. Conclusion

The Random Forest model shows measurable bias against male passengers (under-predicting survival by 6.7%) and 3rd class passengers (under-predicting by 5%). This bias is inherited from historical data rather than introduced by the algorithm. SHAP and LIME clearly demonstrate that `sex` is the dominant decision factor in individual predictions. Responsible deployment of this model would require fairness re-calibration before use in any real-world decision system.

GitHub Repository: <https://github.com/sarthakcoursera/InternSpark-Internship>